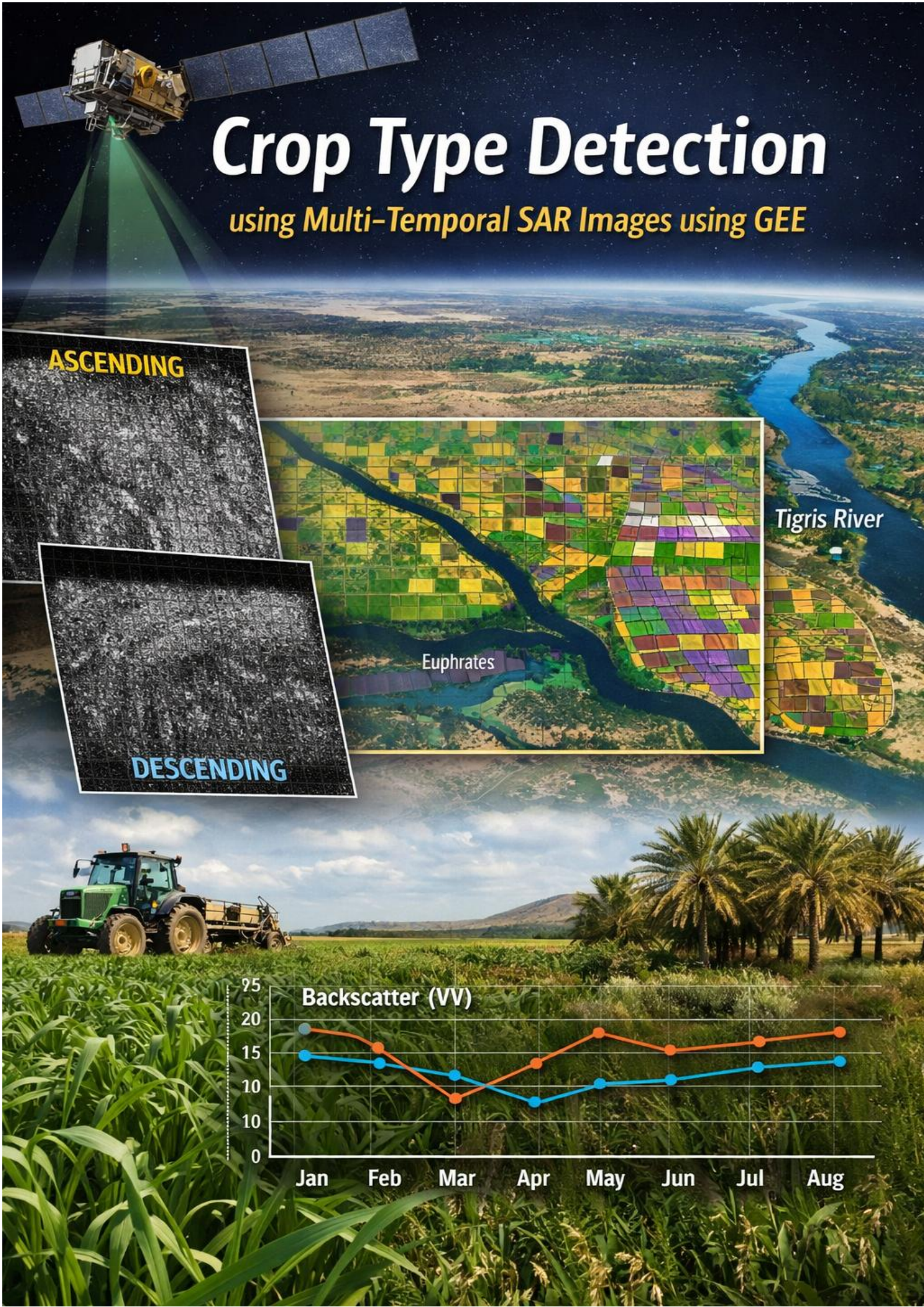
Crop Type Detection

using Multi-Temporal SAR Images using GEE



Abstract

This report presents the complete design, implementation, and scientific rationale of a multi-temporal Sentinel-1 Synthetic Aperture Radar (SAR) time-series monitoring pipeline developed using the Google Earth Engine (GEE) cloud computing platform for crop type detection.

The project focuses on a rectangular area of interest in central Iraq covering approximately 37 km × 27 km and spans the full calendar year 2025.

The pipeline ingests Sentinel-1 C-band SAR Ground Range Detected (GRD) imagery in VV polarisation acquired under the Interferometric Wide (IW) swath mode. Data are processed through five major stages: temporal filtering and polarisation selection, orbit-direction separation into ascending and descending passes, monthly mean compositing, focal mean speckle suppression, and final export as a georeferenced multi-temporal GeoTIFF stack.

The pipeline is designed to support a range of downstream applications including agricultural monitoring, flood extent mapping, land cover classification, and urban growth analysis. All processing is performed entirely server-side within the GEE infrastructure, eliminating the need for local data downloads during development.

Keyword	Value
Sensor	Sentinel-1 A/B · C-band SAR · 5.6 cm wavelength
Product	Ground Range Detected (GRD) · IW mode · VV polarisation
Temporal scope	2025-01-01 to 2025-12-31 (12 monthly composites)
Spatial scope	48.42°31.94° to 48.76°32.19° — ~1,000 km²
Output resolution	30 m (export scale) · EPSG:4326
Platform	Google Earth Engine JavaScript API

1. Introduction

1.1 Background and Motivation

Synthetic Aperture Radar (SAR) remote sensing is a powerful tool for Earth observation, offering significant advantages over optical sensors. Unlike passive systems that rely on sunlight, SAR actively emits microwave signals and measures the returned backscatter, enabling data acquisition regardless of daylight and with minimal interference from clouds, aerosols, or light rain.

Central Iraq, the focus of this study, is a region of major hydrological and agricultural importance. The Tigris and Euphrates river systems support extensive irrigated agriculture, while seasonal flooding, evolving irrigation practices, and urban expansion make the area well-suited for multi-temporal SAR analysis. Optical imagery in this region is often limited by dust storms and atmospheric conditions, further highlighting the value of SAR data.

Google Earth Engine (GEE) facilitates large-scale geospatial analysis by providing cloud-based access to extensive Earth observation datasets and enabling parallel processing through its API. The Sentinel-1 GRD dataset available in GEE offers consistent, multi-year global coverage, making it ideal for time-series monitoring applications such as this study.

1.2 Project Objectives

- Construct an automated, cloud-based pipeline that retrieves, filters, and processes all available Sentinel-1 GRD scenes over the defined AOI.
- Separate data by satellite orbit direction (ascending and descending) to maintain geometric consistency within each composite.
- Generate monthly mean backscatter composites that reduce speckle noise and represent the average surface scattering state for each calendar month.
- Apply spatial speckle filtering to further suppress residual noise before visualization and export.
- Produce a pixel-level time-series chart at a representative point within the AOI for qualitative validation of seasonal backscatter dynamics.
- Export the ascending orbit monthly composite stack to Google Drive as a multi-band GeoTIFF ready for downstream analysis.

2. Scientific Background

2.1 Synthetic Aperture Radar Fundamentals

A SAR (Synthetic Aperture Radar) system transmits microwave pulses toward the Earth and records the returned signal's amplitude and phase. As the satellite moves, it combines echoes from multiple pulses to simulate a very large antenna, enabling high-resolution imaging (meters to tens of meters) from orbit.

GRD (Ground Range Detected) products:
GRD products are SAR images that have been processed and projected into ground range coordinates, with multi-looking applied and radiometric calibration performed. They are typically used for visualization and analysis rather than precise phase measurements.

Errors removed / reduced in GRD:

- **Speckle noise** (reduced via multi-looking/averaging)
- **Geometric distortions in slant range** (converted to ground range projection)
- **Radiometric inconsistencies** (calibrated to provide consistent backscatter values)
- **Range cell ambiguity effects** (partially mitigated through processing)
- **Thermal noise (partially reduced)** depending on processing level

2.2 Sentinel-1 Instrument and Mission

Sentinel-1 is a two-satellite constellation (1A and 1B) operated by the European Space Agency (ESA) as part of the Copernicus Earth Observation Programme. Both satellites carry a C-band SAR instrument operating at 5.405 GHz (wavelength ≈5.6 cm). The constellation provides a combined revisit time of approximately 6 days at the equator, though actual acquisition frequency over any given area depends on the observation scenario and regional priorities.

The Interferometric Wide (IW) swath mode, used exclusively in this project, is the primary acquisition mode over land globally. IW uses the TOPSAR (Terrain Observation by Progressive Scan SAR) technique to acquire data in three sub-swaths with an effective total swath width of approximately 250 km.

2.3 Polarisation and Scattering Mechanisms

Sentinel-1 IW mode transmits in one polarisation and receives in two. In this project, only the VV (vertical transmit and vertical receive) channel is used. VV polarisation is well suited for surface scattering, such as from flat or slightly rough areas like farmland, calm water, and deserts.Surface scattering in these cases is often explained by the Bragg scattering model. It shows that VV backscatter depends strongly on the surface's roughness, moisture (dielectric properties), and the angle at which the radar signal hits the ground.

2.4 Speckle Noise in SAR Imagery

A key limitation of SAR images is speckle noise. Speckle occurs because each pixel contains many small reflection objects (like soil particles, plants, or buildings). The signals from these objects combine in different ways. Sometimes they reinforce each other (making the signal strong), and sometimes they cancel out (making it weak). As a result, even uniform areas appear grainy, giving SAR images a "salt-and-pepper" look, especially in single-look images.

2.5 Ascending vs Descending Orbit Geometry

Sentinel-1 collects data in two orbit directions: ascending (south to north, viewing from the west, usually evening) and descending (north to south, viewing from the east, usually morning). The radar looks sideways at the ground at a fixed angle, so the viewing geometry changes between the two passes.

Because of this, terrain effects like layover (slopes facing the radar appearing compressed) and shadow (areas hidden from the radar) occur on different sides of features in ascending vs descending images. If both are combined, it can create errors in areas with hills or mountains. Also, since the two passes happen at different times of day, they may capture different surface conditions (like moisture or temperature changes), which is another reason to keep them separate.

3. Data and Study Area

3.1 Study Area Description

The area of interest is a rectangular region in central Iraq (approx. 37 km × 27 km, ~1,000 km²) located in the Mesopotamian alluvial lowlands between the Tigris and Euphrates rivers. The terrain is flat to gently undulating, with elevations below 50 m, and land cover includes irrigated agriculture (cereals, vegetables, date palms), bare or fallow soil, small settlements, irrigation canals, and seasonal wetlands. Its flat topography minimizes SAR layover and shadow effects, making it suitable for demonstrating core SAR processing without topographic correction.

3.2 Sentinel-1 GRD Dataset

The project uses the COPERNICUS/S1_GRD Sentinel-1 dataset in Google Earth Engine, which includes pre-processed Level-1 GRD products (thermal noise removal, radiometric calibration to σ°, and terrain correction using SRTM DEM). Backscatter values are stored in linear scale, so processing and compositing are done in linear units; conversion to dB is only applied for visualization using 10·log10(value), as averaging should be performed in linear space rather than in dB.

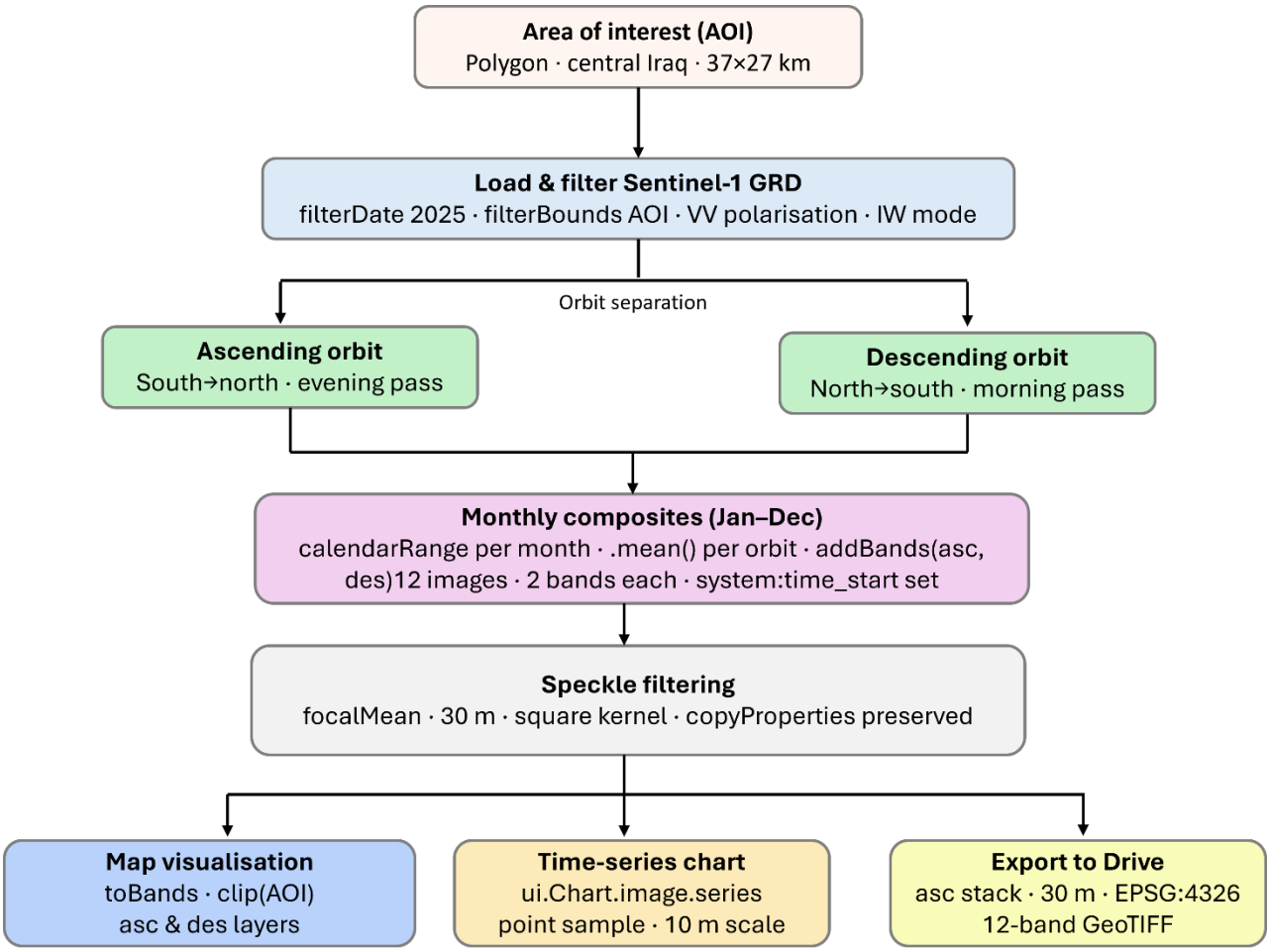
3.3 Temporal Coverage

The analysis covers the full year 2025 (2025-01-01 to 2025-12-31), using GEE's filterDate to select all Sentinel-1 scenes acquired within this period. With the Sentinel-1A and 1B constellation, the AOI typically receives about 2–4 acquisitions per month per orbit direction, resulting in roughly 24–48 ascending and descending scenes combined over the year.

Parameter	Value	Rationale	Impact
Dataset	COPERNICUS/S1_GRD	ESA standard product; radiometrically calibrated	Ready for quantitative analysis
Band selected	VV	Sensitive to soil moisture & surface roughness	Relevant for agricultural & flood monitoring
Instrument mode	IW (TOPSAR)	250 km swath; primary land acquisition mode	Guaranteed AOI coverage each revisit
Time range	2025 full year	Annual cycle captures all seasonal states	12-month compositing possible
Orbit	Both (separated)	Different look geometry = independent information	Doubled temporal density; geometric consistency

4. Methodology

The methodology follows a sequential server-side processing pipeline implemented entirely in the Google Earth Engine JavaScript API.



- This workflow begins Sentinel-1 SAR data in Google Earth Engine for a defined AOI in central Iraq over the year 2025. It begins by defining the rectangular Area of Interest using a polygon geometry, which is used to clip and filter all subsequent data.

- The Sentinel-1 GRD collection is then filtered by date (full year 2025), spatial overlap with the AOI, VV polarisation, and IW acquisition mode, ensuring only relevant SAR scenes are used.
- Next, the data is split into ascending and descending orbit passes to avoid mixing observations with different viewing geometries.

- For each month, separate mean composites are generated for ascending and descending data, producing 12 monthly values per orbit direction.

- These monthly composites are then lightly smoothed using a spatial focal mean filter to reduce residual speckle noise while preserving spatial patterns.

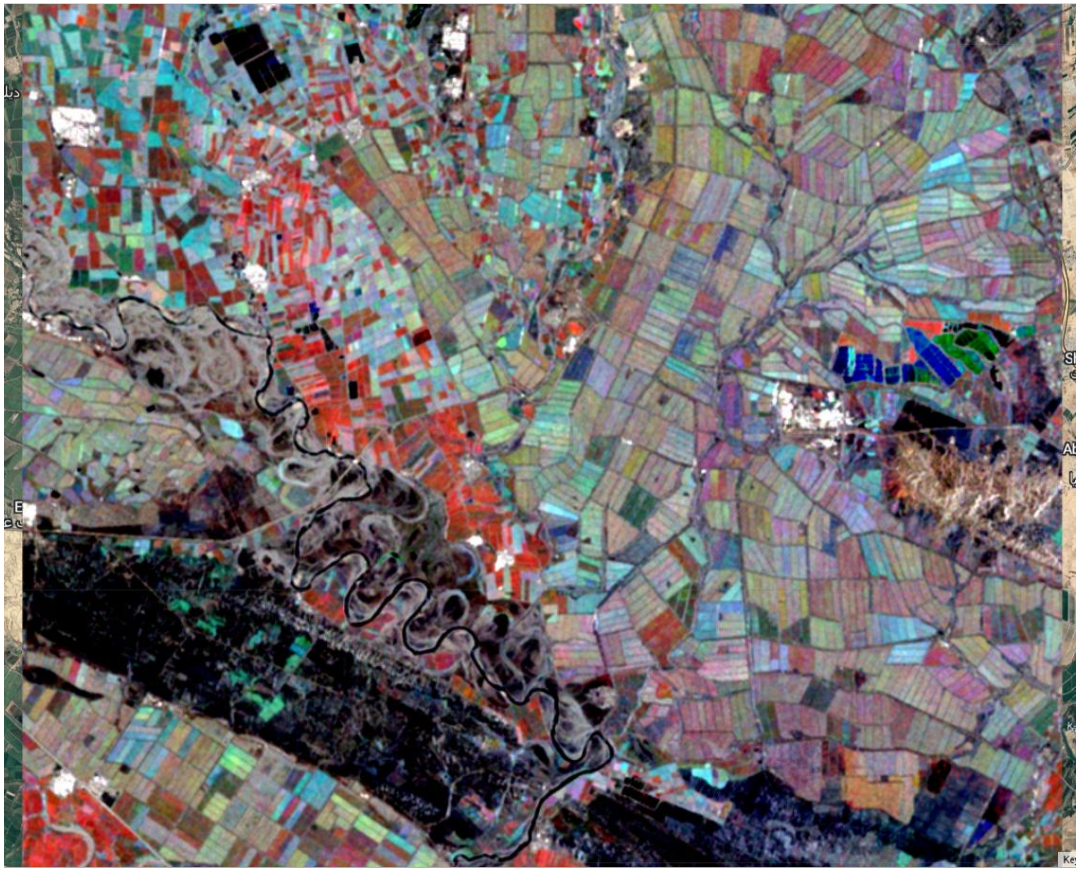
- The results are visualised on the map and sampled at a central point to generate a time-series chart showing backscatter variation over time. Finally, the processed monthly ascending composites are exported as a multi-band GeoTIFF to Google Drive, with each band representing one month of averaged VV backscatter in linear scale, ready for use in GIS or further analysis.

5. Results and Expected Outputs

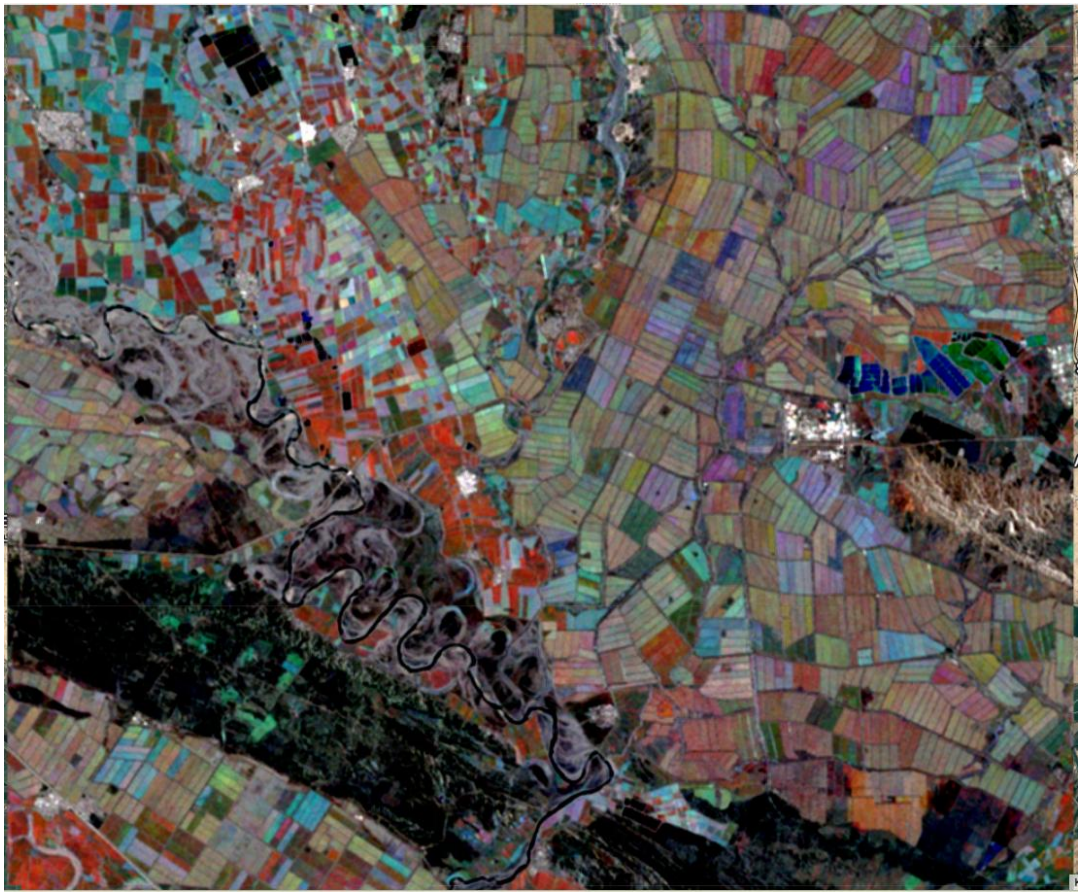
5.1 Monthly Composite Stack

The primary output of the pipeline is a 12-band GeoTIFF file named sar_asc.tif stored in the test folder of the user's Google Drive. Each band represents the spatial mean of all Sentinel-1 ascending orbit VV backscatter observations acquired within that calendar month over the AOI, after application of a 30 m focal mean speckle filter. Band values are in linear power scale (σ° linear). The file has the following specifications:

Property	Value
File format	GeoTIFF (.tif)
Coordinate system	WGS-84 geographic (EPSG:4326)
Pixel resolution	30 m (at equator)
Number of bands	12 (one per calendar month, Jan–Dec 2025)
Band data type	Float32
Band naming	2025-01_asc, 2025-02_asc, …, 2025-12_asc
Backscatter scale	Linear power scale (σ°); convert to dB: $10 \times \log_{10}(\text{value})$
No-data value	GEE default masked pixels
Approximate file size	~50–150 MB depending on pixel count and compression



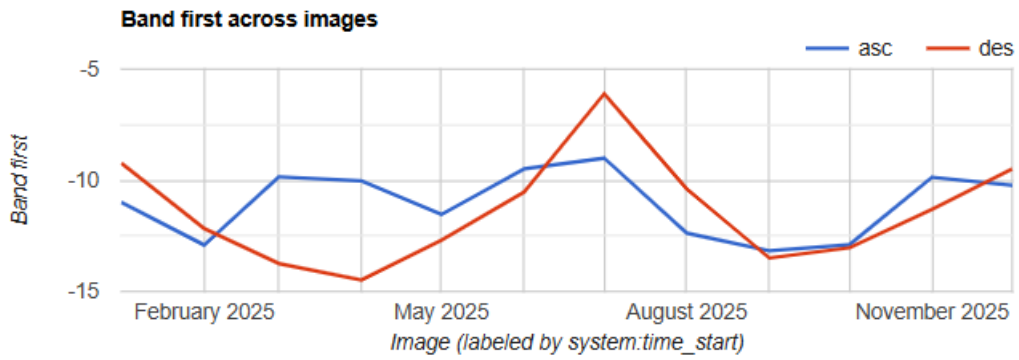
Crop classification for descending orbit-2025



Crop classification for ascending orbit-2025

6.2 Temporal Backscatter Time-Series

The ui.Chart.image.series chart visualizes the monthly mean backscatter at the sample point [48.53°E, 32.08°N] for both orbit directions across the year. In an agricultural landscape, expected seasonal patterns include elevated backscatter in winter and spring months coinciding with crop growth (volume scattering from crop canopies) and reduced backscatter in summer months corresponding to bare, dry post-harvest fields (surface scattering from smooth, dry soil).



6.3 Downstream Applications

The exported multi-temporal SAR stack supports a wide range of downstream analyses. The following examples illustrate common workflows that can be applied to the sar_asc.tif output.

- Agricultural field boundary delineation: Temporal variance of backscatter highlights actively cultivated parcels against stable non-agricultural background, enabling semi-automated field boundary extraction.
- Crop type classification: The 12-band temporal signature constitutes a feature vector for each pixel. Supervised classifiers (Random Forest, SVM) trained on field-surveyed crop type samples can map agricultural land cover.
- Flood mapping: Threshold-based detection of monthly composites with anomalously low VV backscatter identifies potentially inundated areas when validated against pre-flood baseline composites.
- Change detection: Differencing adjacent monthly composites or computing the temporal standard deviation across all 12 bands, highlights pixels with high temporal variability indicative of land use change.
- Urban extent mapping: Persistent high-backscatter areas across all 12 months can be used to delineate urban built-up extents.

7. Limitations and Recommended Improvements

7.1 Current Limitations

The pipeline as implemented has several limitations that should be understood before using outputs for quantitative analysis.

1. Linear-scale averaging: Values are averaged in linear power scale, which is physically correct. However, for visualization, users must convert to dB ($10 \times \log_{10}(x)$) before applying standard SAR colour ramps. The export does not automatically perform this conversion.
2. focalMean is edge-blurring: The 30 m focal mean smooths boundaries between contrasting land covers (e.g., field edges adjacent to canals). Applications requiring sharp boundary delineation should apply the export before speckle filtering, or use an edge-preserving filter such as refined Lee or Gamma-MAP.
3. No geometric terrain correction: The GRD product in GEE includes SRTM-based terrain correction during pre-processing, but the flat terrain of the AOI means this is generally adequate. For sites with even moderate relief, additional Range Doppler Terrain Correction using a high-resolution DEM would be required.
4. Single polarisation: Only VV is used. Including VH and computing the VH/VV ratio would add volume-scattering sensitivity valuable for crop type discrimination and biomass estimation.
5. Monthly compositing hides sub-monthly events: Flood events lasting only a few days may be diluted in a monthly mean composite. For operational flood monitoring, weekly or bi-weekly compositing would be more appropriate.

7.2 Recommended Improvements

1. Add VH polariyation
2. dB converstion before export
3. Implement a Refined Lee or Gamma-MAP filter in GEE to preserve field boundaries and linear features during speckle suppression.
4. After separate compositing, compute the mean of ascending and descending composites to produce a higher-density temporal stack with a look from both directions.

References

Inspired by Google Earth Engine with Amirhossein Ahrari